

Module 3: Basic Statistics — Answer Key

Prequiz

1. For 2, 3, 2, 4, 9:

$$\bar{x} = \frac{20}{5} = \boxed{4},$$

ordered data are 2, 2, 3, 4, 9, so median = $\boxed{3}$, mode = $\boxed{2}$, and range = $9 - 2 = \boxed{7}$.

2. Removing 9 leaves 2, 2, 3, 4:

$$\bar{x} = \frac{11}{4} = \boxed{2.75}, \quad \text{median} = \frac{2+3}{2} = \boxed{2.5}.$$

The mean changed by 1.25, while the median changed by 0.5, so the **mean changed more**.

3. Five students with mean 80 have total score

$$5(80) = 400.$$

The four known scores total $72 + 78 + 85 + 81 = 316$, so the missing score is

$$400 - 316 = \boxed{84}.$$

4. Mean = 5. Squared deviations:

$$(3 - 5)^2 = 4, \quad (5 - 5)^2 = 0, \quad (5 - 5)^2 = 0, \quad (7 - 5)^2 = 4.$$

Their sum is 8, the mean squared deviation is $8/4 = 2$, and

$$s = \sqrt{2} \approx \boxed{1.41}.$$

5. Machine A range: $252 - 248 = \boxed{4}$. Machine B range: $256 - 244 = \boxed{12}$. Machine A is more consistent because its values have less spread.

Matching quiz

- For 6, 7, 6, 8, 13: mean = $40/5 = \boxed{8}$, median = $\boxed{7}$, mode = $\boxed{6}$, range = $13 - 6 = \boxed{7}$.
- Removing 13 leaves 6, 6, 7, 8: mean = $27/4 = \boxed{6.75}$, median = $(6 + 7)/2 = \boxed{6.5}$. The mean changes more.
- Six scores with mean 75 total 450. The five known scores total 370, so the missing score is $\boxed{80}$.
- Mean = 4. Squared deviations are 4, 0, 0, 4, so

$$s = \sqrt{\frac{8}{4}} = \sqrt{2} \approx \boxed{1.41}.$$

5. Set A is more consistent. Both sets have mean 50, but A stays within 49 to 51 while B ranges from 30 to 70.